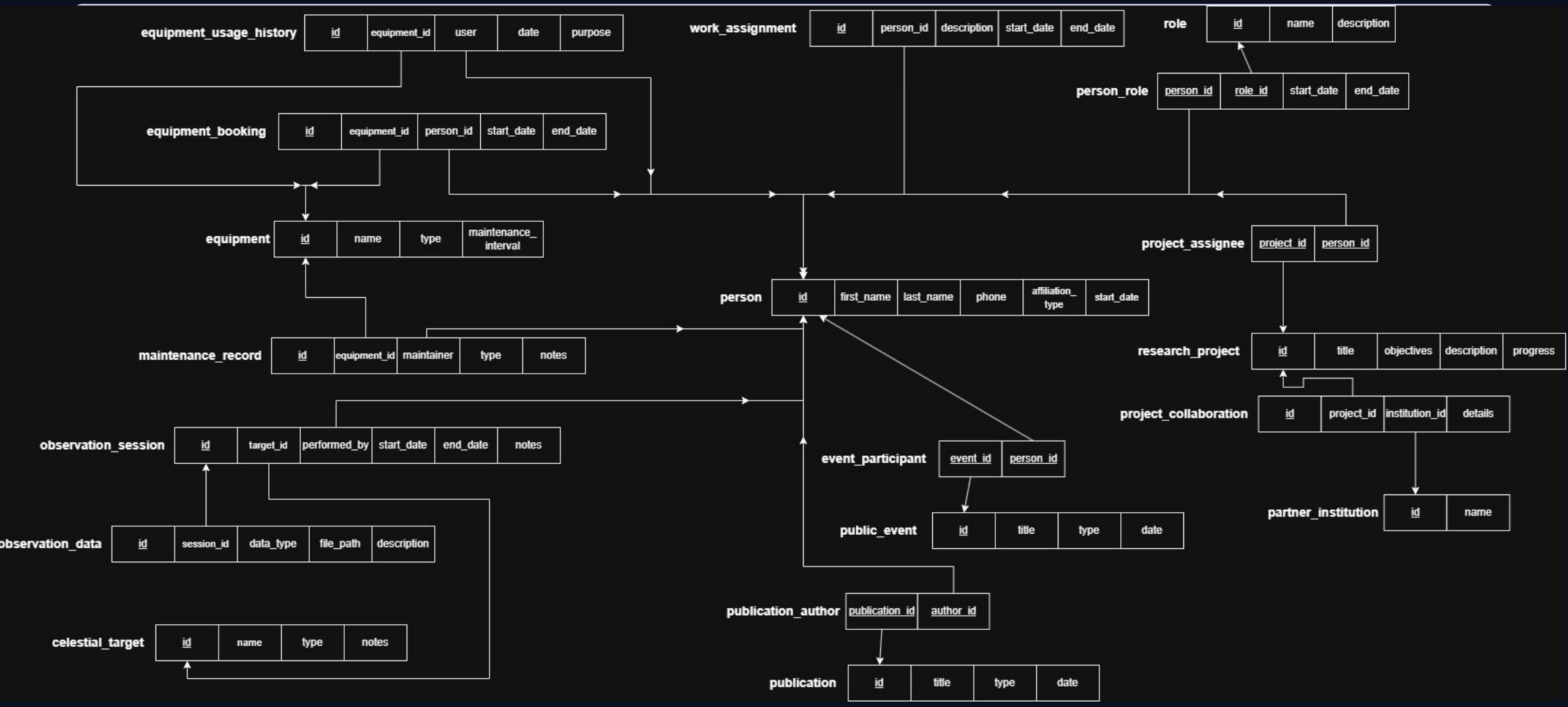
StellarView Management System (SVMS)

StellarView Astronomy Research Center is an organization dedicated to astronomy research, space observation, public education, and maintaining astronomical data. The organization studies celestial objects and events while also helping students, researchers, and the public learn more about the universe.

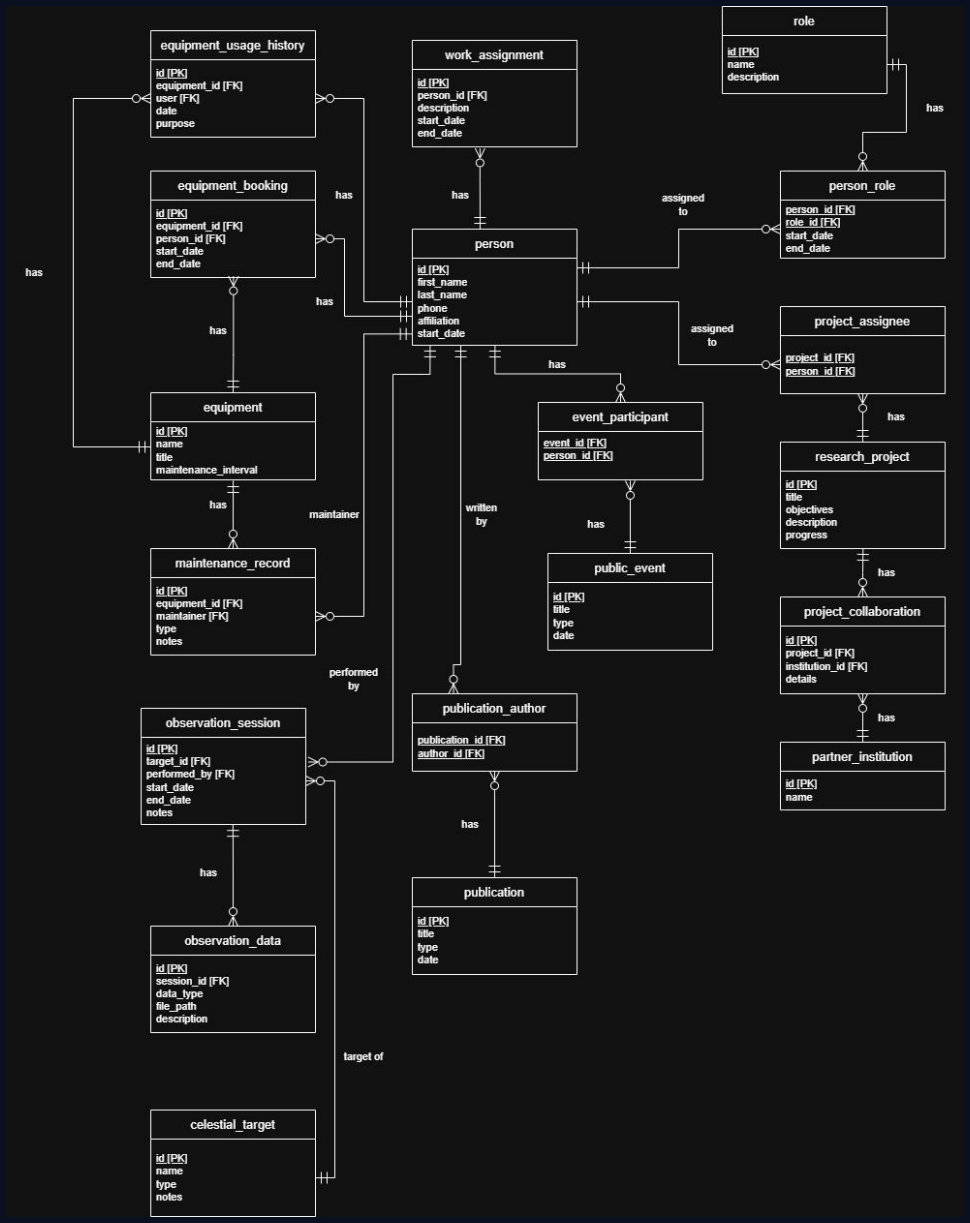
Modules Built

- Research Project Management
- Observation Scheduling
- Astronomical Data Management
- Equipment Management
- Event and Education
- Staff and Researcher Management
- Collaboration and Publication Module

2.) Final Schema Diagram



3.) ER diagram (Crow’s Foot notation)



4.) SQL Database Screenshots

person

id	first_name	last_name	phone	affiliation_type	start_date
1	Amelia	Hart	989-555-0101	Staff	2023-01-10
2	Noah	Bennett	989-555-0102	Researcher	2023-02-14
3	Olivia	Parker	989-555-0103	Researcher	2023-03-01
4	Liam	Foster	989-555-0104	Staff	2023-03-20
5	Emma	Reed	989-555-0105	Student	2023-09-05
6	Mason	Cole	989-555-0106	Student	2023-09-05
7	Sophia	Ward	989-555-0107	Volunteer	2024-01-12
8	James	Brooks	989-555-0108	Staff	2024-02-08
9	Isabella	Hayes	989-555-0109	Researcher	2024-02-15
10	Ethan	Price	989-555-0110	Volunteer	2024-03-01

equipment

id	name	type	maintenance_interval
1	Celestron EdgeHD 11	Telescope	180
2	SkyWatcher EQ6-R Pro	Mount	120
3	ZWO ASI2600MC Pro	Camera	90
4	SBIG Spectrograph Unit	Sensor	150
5	Explore Scientific 2x Barlow	Lens	365
6	Dell Precision Observatory PC	Computer	180
7	Orion StarShoot Autoguider	Camera	120
8	Meade LX200 10	Telescope	180
9	QHYCCD Monochrome Sensor	Sensor	120
10	Portable Outreach Refractor	Telescope	240

role

id	name	description
1	Director	Oversees the overall astronomy program and operati...
2	Lead Researcher	Leads scientific research projects and mentors tea...
3	Telescope Operator	Operates and configures telescopes for scheduled o...
4	Data Analyst	Processes captured measurements, spectra, and imag...
5	Equipment Technician	Maintains observatory equipment and performs calib...
6	Event Coordinator	Plans and manages public workshops and outreach ev...
7	Student Researcher	Supports research projects while learning astronom...
8	Volunteer	Assists with events, logistics, and general suppor...
9	Publication Editor	Prepares reports and research papers for publicati...
10	Collaboration Liaison	Coordinates with partner institutions and shared i...

5.) Top MySQL Queries (1)

1.) Find the title of the public event that had the most participants

```
SELECT pe.title
FROM public_event pe
WHERE
    (
        SELECT COUNT(*)
        FROM event_participant ep
        WHERE ep.event_id = pe.id
    ) =
    (
        SELECT MAX(event_count)
        FROM
        (
            SELECT COUNT(*) as
event_count
            FROM event_participant epo
            GROUP BY epo.event_id
        ) counts
    );
```

Result

title
Filter...
Youth Astronomy Camp Day 2

2.) What is the title and progress of the research project that has the least progress?

```
SELECT rp.title, rp.progress
FROM research_project rp
WHERE rp.progress = (
    SELECT MIN(rpo.progress)
    FROM research_project rpo
);
```

Result

title	progress
Filter...	Filter...
Comet Tail Evolution	29.40

6.) Top MySQL Queries (2)

3.) Who booked the most equipment overall?

```
SELECT p.first_name, p.last_name
FROM person p
WHERE
    (
        SELECT COUNT(*)
        FROM equipment_booking eb
        WHERE eb.person_id = p.id
    ) =
    (
        SELECT MAX(count)
        FROM
        (
            SELECT COUNT(*) as count
            FROM equipment_booking ebo
            GROUP BY ebo.person_id
        ) all_counts
    );
```

Result

first_name	last_name
Filter...	Filter...
Noah	Bennett

4.) What is the average participant count of public events that had at least 1 student participant?

```
SELECT AVG(count)
FROM
(
    SELECT COUNT(*) as count
    FROM public_event pe, event_participant ep, person p
    WHERE pe.id = ep.event_id
        AND p.id = ep.person_id
        AND p.affiliation_type = 'Student'
    GROUP BY pe.id
) all_counts;
```

Result

AVG(count)
Filter...
2.0000

7.) MySQL Challenges (Rigid Schema)

Rigid Schema

- Many tables contain attributes that pertain to a subset of the rows. For example, students and staff are both stored in the person table. Staff require a start_date attribute. Because of this, students also have a start_date attribute that isn't needed

- The opposite problem is also present: we can't add extra attributes to a table without adding that attribute to all records. For example, the celestial_target table stores a variety of celestial objects that have a variety of classification metrics (spectral type, etc). Unfortunately, we cannot store those metrics without somehow adding them to all existing records (with empty or null data).

MongoDB's Flexible Schema

- In MongoDB, it is not required that every record (document) in a table (collection) has all of the same attributes (fields). In a MongoDB design we can still keep the "person" collection, but we'll only add "start date" to the documents that require it (staff, etc).

- MongoDB allows us to add additional fields to any document without requiring every document in that collection to use them. So we can simply add whatever field we want to the relevant documents. For example, we can add the "spectral type" field to only stars.

8.) MySQL Challenges (Nested Lists)

Nested Lists

- Lists tend to be a pain in SQL, as each 1:N relationships requires a table to associate record N with whatever data we're associating it with. For example, the table publication_author associates a publication with an author (person).

- Going further, it's difficult to nest lists within other lists. For example, if the public events had "sub-events" such that the sub-events were not main events, but still had all the same requirements of events (such as tracking participation), then in SQL we'd have to somehow modify our existing schema to satisfy that requirement. We could choose to, for example, add a field for "parent_event_id", if it's not null then we'll know an event is a sub-event

MongoDB's Embedded Structure

- MongoDB allows us to directly place objects inside objects and arrays inside a document. Using this pattern, we can easily nest sub-events to their parent event. This reduces the number of tables (collections).

- Further, we can nest sub-events within sub-events. This is an example where MongoDB encapsulates data much better than SQL. In SQL, every record would have to "know" that the concept of "sub-events" exists (because every record has some sort of "sub-event" attribute). In MongoDB, however, the only records (documents) that know about "sub-events" are the records that use them, and no additional tables are required.

9.) MongoDB Collections

equipment

```
_id: 1
name : "Celestron EdgeHD 11"
type : "Telescope"
maintenance_interval : 180
▸ bookings : Array (1)
▸ maintenance_records : Array (1)
▸ usage_history : Array (1)

_id: 2
name : "SkyWatcher EQ6-R Pro"
type : "Mount"
maintenance_interval : 120
▸ bookings : Array (1)
▸ maintenance_records : Array (1)
▸ usage_history : Array (1)

_id: 3
name : "ZWO ASI2600MC Pro"
type : "Camera"
maintenance_interval : 90
▸ bookings : Array (1)
▸ maintenance_records : Array (1)
▸ usage_history : Array (1)
```

people

```
_id: 1
first_name : "Amelia"
last_name : "Hart"
phone : "989-555-0101"
affiliation_type : "Staff"
start_date : "2023-01-10T00:00:00Z"
▸ roles : Array (1)
▸ work_assignments : Array (3)

_id: 2
first_name : "Noah"
last_name : "Bennett"
phone : "989-555-0102"
affiliation_type : "Researcher"
start_date : "2023-02-14T00:00:00Z"
▸ roles : Array (1)
▸ work_assignments : Array (1)

_id: 3
first_name : "Olivia"
last_name : "Parker"
phone : "989-555-0103"
affiliation_type : "Researcher"
start_date : "2023-03-01T00:00:00Z"
▸ roles : Array (1)
▸ work_assignments : Array (empty)
```

roles

```
_id: 1
name : "Director"
description : "Oversees the overall astronomy program and operations."

_id: 2
name : "Lead Researcher"
description : "Leads scientific research projects and mentors team members."

_id: 3
name : "Telescope Operator"
description : "Operates and configures telescopes for scheduled observations."

_id: 4
name : "Data Analyst"
description : "Processes captured measurements, spectra, and image data."

_id: 5
name : "Equipment Technician"
description : "Maintains observatory equipment and performs calibrations."
```

10.) MongoDB Queries and Results

1.) Find the title of the public event that had the most participants

```
db.public_events.aggregate([
  {
    $project: {
      _id: 0,
      title: 1,
      participant_count: { $size:
"$participants" }
    }
  },
  {
    $sort: { participant_count: -1 }
  },
  {
    $limit: 1
  },
  { $project: { _id: 0, title: 1 } }
]);
```

Result

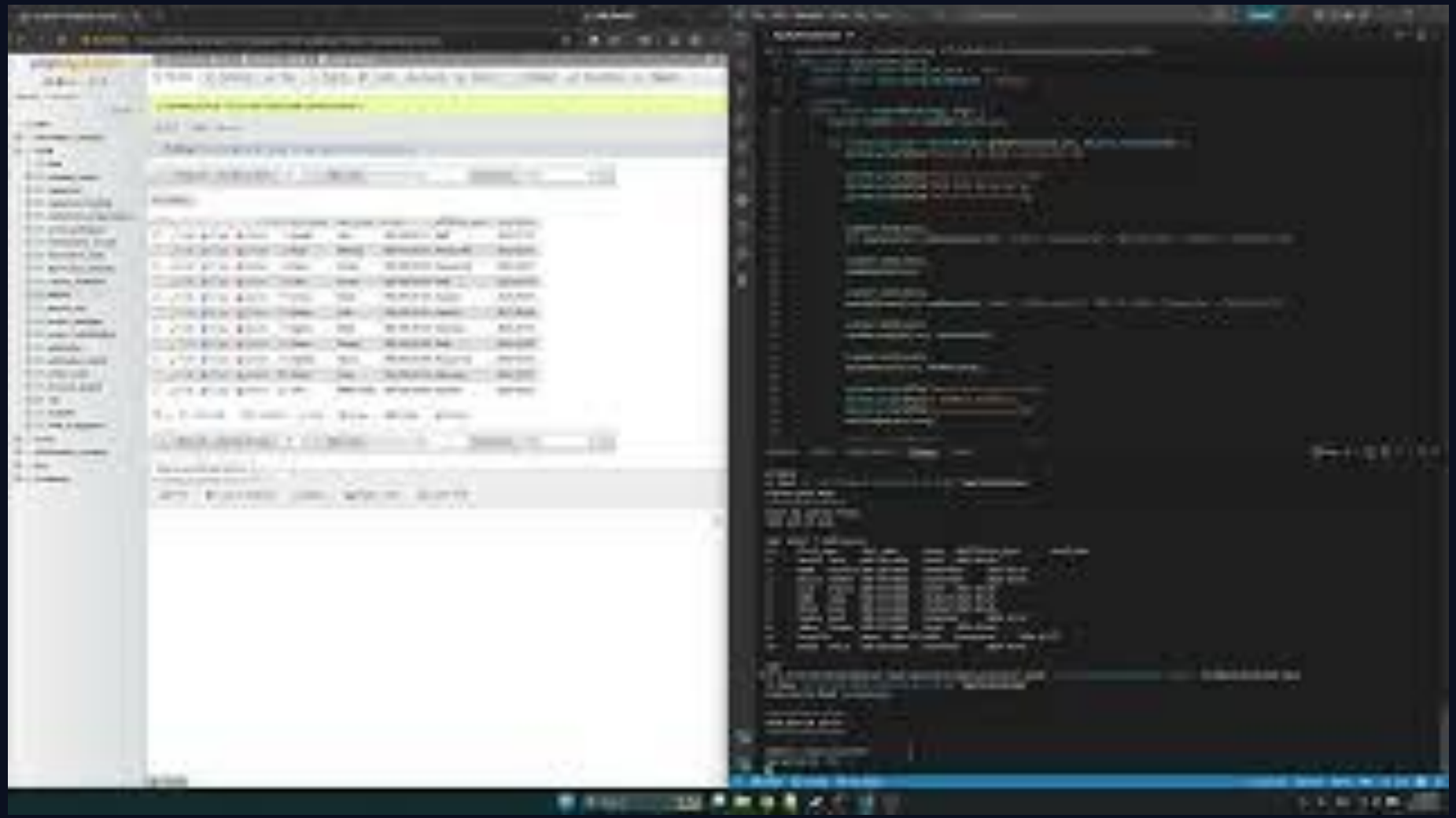
```
< {
    title: 'Youth Astronomy Camp Day 2'
  }
```

2.) What is the title and progress of the research project that has the least progress?

```
db.research_projects.find(
  {},
  {_id: 0, title: 1, progress: 1 }
).sort({ progress: 1 }).limit(1);
```

Result

```
> db.research_projects.find(
    {},
    {_id: 0, title: 1, progress: 1 }
  ).sort({ progress: 1 }).limit(1);
< {
    title: 'Comet Tail Evolution',
    progress: 29.4
  }
```

Application Layer Demo

https://youtu.be/lBwYoF78Pbk